

29.1 Basic configuration

The ADC is configured using the following registers:

1. Power: In the PCONP register ([Table 46](#)), set the PCADC bit.
Remark: On reset, the ADC is disabled. To enable the ADC, first set the PCADC bit, and then enable the ADC in the AD0CR register (bit PDN [Table 532](#)). To disable the ADC, first clear the PDN bit, and then clear the PCADC bit.
2. Clock: In the PCLKSEL0 register ([Table 40](#)), select PCLK_ADC. To scale the clock for the ADC, see bits CLKDIV in [Table 532](#).
3. Pins: Enable ADC0 pins through PINSEL registers. Select the pin modes for the port pins with ADC0 functions through the PINMODE registers ([Section 8.5](#)).
4. Interrupts: To enable interrupts in the ADC, see [Table 536](#). Interrupts are enabled in the NVIC using the appropriate Interrupt Set Enable register. Disable the ADC interrupt in the NVIC using the appropriate Interrupt Set Enable register.
5. DMA: See [Section 29.6.4](#). For GPDMA system connections, see [Table 544](#).

29.2 Features

- 12-bit successive approximation analog to digital converter.
- Input multiplexing among 8 pins.
- Power-down mode.
- Measurement range V_{REFN} to V_{REFP} (typically 3 V; not to exceed V_{DDA} voltage level).
- 12-bit conversion rate of 200 kHz.
- Burst conversion mode for single or multiple inputs.
- Optional conversion on transition on input pin or Timer Match signal.

29.3 Description

Basic clocking for the A/D converters is provided by the APB clock. A programmable divider is included in each converter to scale this clock to the clock (maximum 13 MHz) needed by the successive approximation process. A non-burst mode conversion requires 65 clocks and a burst mode conversion requires 64 clocks.

29.4 Pin description

[Table 530](#) gives a brief summary of each of ADC related pins.

Table 530. ADC pin description

Pin	Type	Description
AD0.7 to AD0.0	Input	Analog Inputs. The ADC cell can measure the voltage on any of these input signals. Digital signals are disconnected from the ADC input pins when the ADC function is selected on that pin in the Pin Select register. Warning: if the ADC is used, signal levels on analog input pins must not be above the level of V_{DDA} at any time. Otherwise, A/D converter readings will be invalid. If the A/D converter is not used in an application then the pins associated with A/D inputs can be used as 5 V tolerant digital IO pins.
V_{REFP} , V_{REFN}	Reference	Voltage References. These pins provide a voltage reference level for the ADC and DAC. Note: V_{REFP} should be tied to VDD(3V3) and V_{REFN} should be tied to V_{SS} if the ADC and DAC are not used.
V_{DDA} , V_{SSA}	Power	Analog Power and Ground. These should typically be the same voltages as V_{DD} and V_{SS} , but should be isolated to minimize noise and error. Note: V_{DDA} should be tied to VDD(3V3) and V_{SSA} should be tied to VSS if the ADC and DAC are not used.

29.5 Register description

The A/D Converter registers are shown in [Table 531](#).

Table 531. ADC registers

Generic Name	Description	Access	Reset value ^[1]	AD0 Name & Address
ADCR	A/D Control Register. The ADCR register must be written to select the operating mode before A/D conversion can occur.	R/W	1	AD0CR - 0x4003 4000
ADGDR	A/D Global Data Register. This register contains the ADC's DONE bit and the result of the most recent A/D conversion.	R/W	NA	AD0GDR - 0x4003 4004
ADINTEN	A/D Interrupt Enable Register. This register contains enable bits that allow the DONE flag of each A/D channel to be included or excluded from contributing to the generation of an A/D interrupt.	R/W	0x100	AD0INTEN - 0x4003 400C
ADDR0	A/D Channel 0 Data Register. This register contains the result of the most recent conversion completed on channel 0.	RO	NA	AD0DR0 - 0x4003 4010
ADDR1	A/D Channel 1 Data Register. This register contains the result of the most recent conversion completed on channel 1.	RO	NA	AD0DR1 - 0x4003 4014
ADDR2	A/D Channel 2 Data Register. This register contains the result of the most recent conversion completed on channel 2.	RO	NA	AD0DR2 - 0x4003 4018
ADDR3	A/D Channel 3 Data Register. This register contains the result of the most recent conversion completed on channel 3.	RO	NA	AD0DR3 - 0x4003 401C
ADDR4	A/D Channel 4 Data Register. This register contains the result of the most recent conversion completed on channel 4.	RO	NA	AD0DR4 - 0x4003 4020
ADDR5	A/D Channel 5 Data Register. This register contains the result of the most recent conversion completed on channel 5.	RO	NA	AD0DR5 - 0x4003 4024
ADDR6	A/D Channel 6 Data Register. This register contains the result of the most recent conversion completed on channel 6.	RO	NA	AD0DR6 - 0x4003 4028
ADDR7	A/D Channel 7 Data Register. This register contains the result of the most recent conversion completed on channel 7.	RO	NA	AD0DR7 - 0x4003 402C
ADSTAT	A/D Status Register. This register contains DONE and OVERRUN flags for all of the A/D channels, as well as the A/D interrupt/DMA flag.	RO	0	AD0STAT - 0x4003 4030
ADTRM	ADC trim register.	R/W	0x0000 0F00	AD0TRM - 0x4003 4034

[1] Reset value reflects the data stored in used bits only. It does not include reserved bits content.

29.5.1 A/D Control Register (AD0CR - 0x4003 4000)

Table 532: A/D Control Register (AD0CR - address 0x4003 4000) bit description

Bit	Symbol	Value	Description	Reset value
7:0	SEL		Selects which of the AD0.7:0 pins is (are) to be sampled and converted. For AD0, bit 0 selects Pin AD0.0, and bit 7 selects pin AD0.7. In software-controlled mode, only one of these bits should be 1. In hardware scan mode, any value containing 1 to 8 ones is allowed. All zeroes is equivalent to 0x01.	0x01
15:8	CLKDIV		The APB clock (PCLK_ADC0) is divided by (this value plus one) to produce the clock for the A/D converter, which should be less than or equal to 13 MHz. Typically, software should program the smallest value in this field that yields a clock of 13 MHz or slightly less, but in certain cases (such as a high-impedance analog source) a slower clock may be desirable.	0
16	BURST	1	The AD converter does repeated conversions at up to 200 kHz, scanning (if necessary) through the pins selected by bits set to ones in the SEL field. The first conversion after the start corresponds to the least-significant 1 in the SEL field, then higher numbered 1-bits (pins) if applicable. Repeated conversions can be terminated by clearing this bit, but the conversion that's in progress when this bit is cleared will be completed. Remark: START bits must be 000 when BURST = 1 or conversions will not start. If BURST is set to 1, the ADGINTEN bit in the AD0INTEN register (Table 534) must be set to 0.	0
		0	Conversions are software controlled and require 65 clocks.	
20:17	-		Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA
21	PDN	1	The A/D converter is operational.	0
		0	The A/D converter is in power-down mode.	
23:22	-		Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA
26:24	START		When the BURST bit is 0, these bits control whether and when an A/D conversion is started:	0
		000	No start (this value should be used when clearing PDN to 0).	
		001	Start conversion now.	
		010	Start conversion when the edge selected by bit 27 occurs on the P2.10 / EINT0 / NMI pin. Set the pin function to EINT0 in PINSEL4 register.	
		011	Start conversion when the edge selected by bit 27 occurs on the P1.27 / CLKOUT / USB_OVRcRn / CAP0.1 pin. Set the pin function to CAP0.1 in PINSEL3 register.	
		100	Start conversion when the edge selected by bit 27 occurs on MAT0.1. Note that this does not require that the MAT0.1 function appear on a device pin.	
		101	Start conversion when the edge selected by bit 27 occurs on MAT0.3. Note that it is not possible to cause the MAT0.3 function to appear on a device pin.	
		110	Start conversion when the edge selected by bit 27 occurs on MAT1.0. Note that this does not require that the MAT1.0 function appear on a device pin.	
		111	Start conversion when the edge selected by bit 27 occurs on MAT1.1. Note that this does not require that the MAT1.1 function appear on a device pin.	
27	EDGE		This bit is significant only when the START field contains 010-111. In these cases:	0
		1	Start conversion on a falling edge on the selected CAP/MAT signal.	
		0	Start conversion on a rising edge on the selected CAP/MAT signal.	
31:28	-		Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA

29.5.2 A/D Global Data Register (AD0GDR - 0x4003 4004)

The A/D Global Data Register holds the result of the most recent A/D conversion that has completed, and also includes copies of the status flags that go with that conversion.

Results of ADC conversion can be read in one of two ways. One is to use the A/D Global Data Register to read all data from the ADC. Another is to use the A/D Channel Data Registers. It is important to use one method consistently because the DONE and OVERRUN flags can otherwise get out of synch between the AD0GDR and the A/D Channel Data Registers, potentially causing erroneous interrupts or DMA activity.

Table 533: A/D Global Data Register (AD0GDR - address 0x4003 4004) bit description

Bit	Symbol	Description	Reset value
3:0	-	Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA
15:4	RESULT	When DONE is 1, this field contains a binary fraction representing the voltage on the AD0[n] pin selected by the SEL field, as it falls within the range of V_{REFP} to V_{REFN} . Zero in the field indicates that the voltage on the input pin was less than, equal to, or close to that on V_{REFN} , while 0xFFF indicates that the voltage on the input was close to, equal to, or greater than that on V_{REFP} .	NA
23:16	-	Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA
26:24	CHN	These bits contain the channel from which the RESULT bits were converted (e.g. 000 identifies channel 0, 001 channel 1...).	NA
29:27	-	Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA
30	OVERRUN	This bit is 1 in burst mode if the results of one or more conversions was (were) lost and overwritten before the conversion that produced the result in the RESULT bits. This bit is cleared by reading this register.	0
31	DONE	This bit is set to 1 when an A/D conversion completes. It is cleared when this register is read and when the ADCR is written. If the ADCR is written while a conversion is still in progress, this bit is set and a new conversion is started.	0

29.5.3 A/D Interrupt Enable register (AD0INTEN - 0x4003 400C)

This register allows control over which A/D channels generate an interrupt when a conversion is complete. For example, it may be desirable to use some A/D channels to monitor sensors by continuously performing conversions on them. The most recent results are read by the application program whenever they are needed. In this case, an interrupt is not desirable at the end of each conversion for some A/D channels.

Table 534: A/D Interrupt Enable register (AD0INTEN - address 0x4003 400C) bit description

Bit	Symbol	Value	Description	Reset value
0	ADINTEN0	0	Completion of a conversion on ADC channel 0 will not generate an interrupt.	0
		1	Completion of a conversion on ADC channel 0 will generate an interrupt.	
1	ADINTEN1	0	Completion of a conversion on ADC channel 1 will not generate an interrupt.	0
		1	Completion of a conversion on ADC channel 1 will generate an interrupt.	
2	ADINTEN2	0	Completion of a conversion on ADC channel 2 will not generate an interrupt.	0
		1	Completion of a conversion on ADC channel 2 will generate an interrupt.	

Table 534: A/D Interrupt Enable register (AD0INTEN - address 0x4003 400C) bit description ...continued

Bit	Symbol	Value	Description	Reset value
3	ADINTEN3	0	Completion of a conversion on ADC channel 3 will not generate an interrupt.	0
		1	Completion of a conversion on ADC channel 3 will generate an interrupt.	
4	ADINTEN4	0	Completion of a conversion on ADC channel 4 will not generate an interrupt.	0
		1	Completion of a conversion on ADC channel 4 will generate an interrupt.	
5	ADINTEN5	0	Completion of a conversion on ADC channel 5 will not generate an interrupt.	0
		1	Completion of a conversion on ADC channel 5 will generate an interrupt.	
6	ADINTEN6	0	Completion of a conversion on ADC channel 6 will not generate an interrupt.	0
		1	Completion of a conversion on ADC channel 6 will generate an interrupt.	
7	ADINTEN7	0	Completion of a conversion on ADC channel 7 will not generate an interrupt.	0
		1	Completion of a conversion on ADC channel 7 will generate an interrupt.	
8	ADGINTEN	0	Only the individual ADC channels enabled by ADINTEN7:0 will generate interrupts. Remark: This bit must be set to 0 in burst mode (BURST = 1 in the AD0CR register).	1
		1	Only the global DONE flag in ADDR is enabled to generate an interrupt.	
31:17	-		Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA

29.5.4 A/D Data Registers (AD0DR0 to AD0DR7 - 0x4003 4010 to 0x4003 402C)

The A/D Data Registers hold the result of the last conversion for each A/D channel, when an A/D conversion is complete. They also include the flags that indicate when a conversion has been completed and when a conversion overrun has occurred.

Results of ADC conversion can be read in one of two ways. One is to use the A/D Global Data Register to read all data from the ADC. Another is to use the A/D Channel Data Registers. It is important to use one method consistently because the DONE and OVERRUN flags can otherwise get out of synch between the AD0GDR and the A/D Channel Data Registers, potentially causing erroneous interrupts or DMA activity.

Table 535: A/D Data Registers (AD0DR0 to AD0DR7 - 0x4003 4010 to 0x4003 402C) bit description

Bit	Symbol	Description	Reset value
3:0	-	Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA
15:4	RESULT	When DONE is 1, this field contains a binary fraction representing the voltage on the AD0[n] pin, as it falls within the range of V_{REFP} to V_{REFN} . Zero in the field indicates that the voltage on the input pin was less than, equal to, or close to that on V_{REFN} , while 0xFFF indicates that the voltage on the input was close to, equal to, or greater than that on V_{REFP} .	NA
29:16	-	Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA
30	OVERRUN	This bit is 1 in burst mode if the results of one or more conversions was (were) lost and overwritten before the conversion that produced the result in the RESULT bits. This bit is cleared by reading this register.	NA
31	DONE	This bit is set to 1 when an A/D conversion completes. It is cleared when this register is read.	NA

29.5.5 A/D Status register (ADSTAT - 0x4003 4030)

The A/D Status register allows checking the status of all A/D channels simultaneously. The DONE and OVERRUN flags appearing in the ADDRn register for each A/D channel are mirrored in ADSTAT. The interrupt flag (the logical OR of all DONE flags) is also found in ADSTAT.

Table 536: A/D Status register (AD0STAT - address 0x4003 4030) bit description

Bit	Symbol	Description	Reset value
0	DONE0	This bit mirrors the DONE status flag from the result register for A/D channel 0.	0
1	DONE1	This bit mirrors the DONE status flag from the result register for A/D channel 1.	0
2	DONE2	This bit mirrors the DONE status flag from the result register for A/D channel 2.	0
3	DONE3	This bit mirrors the DONE status flag from the result register for A/D channel 3.	0
4	DONE4	This bit mirrors the DONE status flag from the result register for A/D channel 4.	0
5	DONE5	This bit mirrors the DONE status flag from the result register for A/D channel 5.	0
6	DONE6	This bit mirrors the DONE status flag from the result register for A/D channel 6.	0
7	DONE7	This bit mirrors the DONE status flag from the result register for A/D channel 7.	0
8	OVERRUN0	This bit mirrors the OVERRRUN status flag from the result register for A/D channel 0.	0
9	OVERRUN1	This bit mirrors the OVERRRUN status flag from the result register for A/D channel 1.	0
10	OVERRUN2	This bit mirrors the OVERRRUN status flag from the result register for A/D channel 2.	0
11	OVERRUN3	This bit mirrors the OVERRRUN status flag from the result register for A/D channel 3.	0
12	OVERRUN4	This bit mirrors the OVERRRUN status flag from the result register for A/D channel 4.	0
13	OVERRUN5	This bit mirrors the OVERRRUN status flag from the result register for A/D channel 5.	0
14	OVERRUN6	This bit mirrors the OVERRRUN status flag from the result register for A/D channel 6.	0
15	OVERRUN7	This bit mirrors the OVERRRUN status flag from the result register for A/D channel 7.	0
16	ADINT	This bit is the A/D interrupt flag. It is one when any of the individual A/D channel Done flags is asserted and enabled to contribute to the A/D interrupt via the ADINTEN register.	0
31:17	-	Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA

29.5.6 A/D Trim register (ADTRIM - 0x4003 4034)

This register will be set by the bootcode on start-up. It contains the trim values for the DAC and the ADC. The offset trim values for the ADC can be overwritten by the user. All 12 bits are visible when this register is read.

Table 537: A/D Trim register (ADTRM - address 0x4003 4034) bit description

Bit	Symbol	Description	Reset value
3:0	-	reserved.	NA
7:4	ADCOFFS	Offset trim bits for ADC operation. Initialized by the boot code. Can be overwritten by the user.	0
11:8	TRIM	written-to by boot code. Can not be overwritten by the user. These bits are locked after boot code write.	1111
31:12	-	Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA

29.6 Operation

Once an ADC conversion is started, it cannot be interrupted. A new software write to launch a new conversion or a new edge-trigger event will be ignored while the previous conversion is in progress.

29.6.1 Hardware-triggered conversion

If the BURST bit in the ADCR is 0 and the START field contains 010-111, the ADC will start a conversion when a transition occurs on a selected pin or Timer Match signal. The choices include conversion on a specified edge of any of 4 Match signals, or conversion on a specified edge of either of 2 Capture/Match pins. The pin state from the selected pad or the selected Match signal, XORed with ADCR bit 27, is used in the edge detection logic.

29.6.2 Interrupts

An interrupt request is asserted to the NVIC when the DONE bit is 1. Software can use the Interrupt Enable bit for the A/D Converter in the NVIC to control whether this assertion results in an interrupt. DONE is negated when the ADDR is read.

29.6.3 Accuracy vs. digital receiver

The ADC function must be selected via the PINSEL registers in order to get accurate voltage readings on the monitored pin. The PINMODE should also be set to the mode for which neither pull-up nor pull-down resistor is enabled. For a pin hosting an ADC input, it is not possible to have a digital function selected and yet get valid ADC readings. An inside circuit disconnects ADC hardware from the associated pin whenever a digital function is selected on that pin.

29.6.4 DMA control

A DMA transfer request is generated from the ADC interrupt request line. To generate a DMA transfer the same conditions must be met as the conditions for generating an interrupt (see [Section 29.6.2](#) and [Section 29.5.3](#)).

Remark: If the DMA is used, the ADC interrupt must be disabled in the NVIC.

For DMA transfers, only burst requests are supported. The burst size can be set to one in the DMA channel control register (see [Section 31.5.20](#)). If the number of ADC channels is not equal to one of the other DMA-supported burst sizes (applicable DMA burst sizes are 1, 4, 8 - see [Section 31.5.20](#)), set the burst size to one.

The DMA transfer size determines when a DMA interrupt is generated. The transfer size can be set to the number of ADC channels being converted (see [Section 31.5.20](#)). Non-contiguous channels can be transferred by the DMA using the scatter/gather linked lists (see [Section 31.5.19](#)).